Cross-Modality Image Matching Network With Modality-Invariant Feature Representation for Airborne-Ground Thermal Infrared and Visible Datasets

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***Abstract*— Thermal infrared (TIR) remote-sensing imagery can allow objects to be imaged clearly at night through the long-wave infrared, so that the fusion of thermal infrared and visible (VIS) imagery is a way to improve the remote-sensing interpretation ability. However, due to the large radiation difference between the two kinds of images, it is very difficult to match them. One of the most important issues is the lack of comprehensive consideration of the modality-specific information and modality-shared information, which makes it difficult for the existing methods to obtain a modality-invariant feature representation. In this article, a cross-modality image matching network, which we refer to as CMM-Net, is proposed to realize thermal infrared and visible image matching by learning a modality-invariant feature representation. First, in order to extract the modality-specific features of the imagery, the framework constructs a shallow two-branch network to make full use of the modality-specific information, without sharing parameters. Second, in order to extract the high-level semantic information between the different modalities, modality-shared layers are embedded into the deep layers of the network. In addition, three novel loss functions are designed and combined to learn the modality-invariant feature representation, that is, the discriminative loss of the non-corresponding features in the same modality, the cross-modality loss of the corresponding features between different modalities, and the cross-modality triplet (CMT) loss. The multimodal matching experiments conducted with ground- and airborne-based thermal infrared images and visible images showed that the proposed method outperforms the existing image matching methods by about 2% and 6% for the ground and airborne images, respectively.**

***Index Terms*— Convolutional neural network (CNN), cross-modality matching, deep learning, image matching, modality-invariant feature representation, thermal infrared (TIR) imagery, visible (VIS) imagery.**

I. Introduction

THERMAL infrared (TIR) imagery can allow us to distinguish the targets from the background according to the thermal radiation difference and conduct imaging throughout the day and night [1]. Visible (VIS) images can provide high-resolution texture details in a manner consistent with the human visual system. The combined application of these two types of images can take advantage of the radiation information of objects in the thermal infrared imagery and the feature contour and texture information in the visible imagery [2]. After matching these two cross-modality image types, their combined application and joint interpretation can make Earth observation more powerful [3]–[5]. However, image matching is challenging due to the heterogeneity of the data, as their signals come from different radiation mechanisms. Visible images capture reflected light in the visible band, while thermal infrared images represent the radiation emitted by the target, which is also known as thermal radiation, and is highly correlated with the temperature of the object [6], [7]. According to the law of thermodynamics, thermal radiation has diffusivity, so that it conducts from a high-temperature object to a low-temperature object. In terms of the image matching task, it is very difficult to extract obvious contours and structures from thermal infrared imagery [8], [9].

The existing image matching methods are based on handcrafted descriptors and are dedicated to extracting geometric features, such as the distinct contours and structures of the images. These methods implement feature matching using selected measures, such as scale-invariant feature transform (SIFT) [10], speeded-up robust features (SURFs) [11], oriented fast and rotated brief (ORB) [12], and KAZE [13]. However, the matching of thermal infrared and visible images based on handcrafted descriptors has two main problems.

1) In the feature extraction stage, the abstraction ability of the handcrafted descriptors is insufficient, and they rely on the low-level information of the image, such as corners, gradients, edges, and so on. However, due to the thermal diffusion effect of thermal infrared imagery, gradient smoothing and edge blurring can result in the failure of the feature extraction in the thermal infrared imagery.
2) In the feature matching stage, the feature similarity measurement criteria (such as the nearest neighbor search [14] or Euclidean distance matching [15]) are

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easily limited to locally optimal solutions. Therefore, the matching effect of handcrafted descriptors applied to thermal infrared and visible images is generally poor.

In recent years, the data-driven approach, as presented by deep learning, has been successfully introduced into the field of image matching [16]–[20]. Several nonlinear data mapping functions can be stacked to form a deep-level network function, which approximates arbitrarily complex functions based on the support of a large number of training samples [21]. The detect-and-describe network (D2-Net) [22] adopts a convolutional neural network (CNN) to generate the feature map from the original imagery, and then feature detection is performed on the feature map. For thermal infrared and visible image matching, the existing methods based on deep learning mainly adopt a two-branch framework, in which the CNN acts as both the feature extractor and feature descriptor [23]–[27]. The two-branch networks can be divided into Siamese networks and pseudo-Siamese networks. Siamese networks use shared parameters to reduce the training cost of the two-branch network [28], but for the cross-modality image matching task, Siamese networks cannot easily describe the difference between the different modalities. The pseudo-Siamese networks, without shared parameters, can be adopted to learn different feature representations according to the different modalities. As a result, pseudo-Siamese networks have proved more suitable than Siamese networks in solving the problems of cross-modality image matching [29], [30]. Through practical research, we found that the visual representation of low-level features with varying modalities is very different in the feature description of a shallow network. The visible imagery describes the edge and texture information of the objects, while the thermal infrared imagery reflects the temperature of the objects. There are significant differences between these image types, so in a shallow network, a pseudo-Siamese network can help capture this difference, which is crucial for the image matching.

However, the use of pseudo-Siamese networks for cross-modality image matching is faced with some challenges.

1) For the design of the network, the mechanism of unshared parameters is not conducive to learning a shared space independent of the modality. However, although the thermal infrared and visible images are cross-modality heterogeneous images, they still have some similar semantic information, such as the shape, structure, and topological relationship. Deep CNNs can extract semantic features at deeper levels, which are shared by the different modalities. In fact, it is precisely this high-level semantic information that human beings rely on when interpreting multimodal images.
2) For the network loss function, there are few loss functions specially designed for cross-modality image matching. The matching of thermal infrared and visible images is a cross-modality problem, and the data may have high intra-modality variability, which makes the cross-modality image matching task extremely challenging. Compared to near-infrared images, thermal infrared images describe the emission energy of ground objects, that is, the temperature information. The embodiment of the image matching task is that the intraclass difference of the extracted features is increased, which reduces the uniqueness and differentiation of the features. The existing deep learning-based image matching methods for infrared images are usually based on near-infrared images [31], [28], rather than thermal infrared images. As a result, they do not focus on the problem of the large intraclass difference of the features, so that the results of applying them to thermal infrared and visible images are often disappointing.

In this article, in order to solve the above problems, a novel network called CMM-Net is proposed. The main contributions of this work are as follows.

A. Novel CMM-Net for Thermal Infrared and Visible Images

Thermal infrared and visible images have great visual differences due to the different band compositions. However, the existing methods lack comprehensive consideration of the modality-specific information and modality-shared information. CMM-Net adopts a pseudo-Siamese network with unshared parameters in the shallow layers to learn a modality-specific feature representation, while embedding parameter-shared layers in the deeper layers of the network to learn a modality-invariant feature representation. CMM-Net makes full use of both the modality-specific information and the modality-shared knowledge, which is valuable for reducing the cross-modality differences and generating invariant features.

B. Modality-Specific Discriminative (MSD) Loss Function for Expanding the Inter-Class Distance to Learn a Modality-Invariant Feature Representation

Although the thermal infrared and visible images are cross-modality heterogeneous images, they still have some similar semantic information, such as the structure or the topological relationships. With the help of a deep convolutional network, CMM-Net learns a common representative space where the data from various modalities can be compared. The MSD loss function maximizes the interclass distance to separate the modality-specific features and modality-invariant features.

C. Cross-Modality Constraint (CMC) With Euclidean Distance to Narrow the Cross-Modality Gap

The modal difference between thermal infrared and visible images leads to the dissimilarity of the representation of corresponding features. In the modality-invariant feature space, it is necessary to impose further constraints to make the distribution of the corresponding features more compact. The CMC reduces the distance between the cross-modality intrafeatures, thereby reducing the differences between the corresponding features from different modalities.

Airborne image datasets obtained by unmanned aerial vehicles (UAVs) and ground image datasets captured by sensors on the ground were used as the experimental data to verify the effectiveness of CMM-Net. The test results obtained on these

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challenging datasets demonstrate that CMM-Net is superior to the current state-of-the-art image matching methods.

The rest of this article is organized as follows. The two-branch networks for image matching are summarized in Section II. Section III describes the proposed CMM-Net, including the framework of CMM-Net, the detection of key features, and the design of the loss functions. In Section IV, we describe how thermal infrared and visible images from airborne and ground platforms were used as the experimental datasets to verify the effectiveness of CMM-Net. Section V introduces the ablation study and sensitivity analysis. Our conclusion is summarized in Section VI.

II. Background

This section describes the two-branch network structures, including the Siamese network structure and the pseudo-Siamese network structure. The classical algorithms in natural image matching and their application in remote sensing image matching are also introduced. The main problems with the two-branch network application in cross-modality image matching are also summarized.

The Siamese network was first proposed by Chopra *et al.* in 2005 [32]. The Siamese network structure contains two branches, in which two images are input at the same time, and the output of the network is the similarity value, the purpose of which is to compare the similarity of the two images. Therefore, this type of matching network can be regarded as a patch-based matching problem. The structure of the network is composed of two main parts.

1) The two feature extraction branches, which extract their feature representation through the CNN. The weights between the two branches are shared to reduce parameters.
2) The classifier component, which generates the matching probability by merging the elements in the two branches. This is a supervised end-to-end framework that implements the two parts as a whole. To reduce the computation, the original Siamese network's dual-branch structure adopts exactly the same parameters. With the increasing application of Siamese networks, the pseudo-Siamese network with unshared parameters was derived to describe the unique features of the two images. Alongside the development of deep learning, Siamese networks have been widely used in face recognition [33]–[36], person re-identification [35], [37], and change detection [38], [39].

Over the past five years, Siamese networks have become one of the most popular network architectures for image matching tasks and have been widely used with natural images [40], [41]. MatchNet successfully transforms the matching problem into a classification problem by connecting the two-branch network for feature description with the measurement network and training with the cross-entropy loss function [42]. DeepCompare explores a variety of different neural network structures, namely Siamese networks, pseudo-Siamese networks, two-channel networks, spatial pyramid pooling (SPP) Siamese networks, as well as two-stream multiresolution models called central-surround two-stream networks [43]. DeepDesc learns 128-dimensional descriptors whose Euclidean distances reflect the patch similarity [44]. pn-Net proposes the use of a softened version of triplet loss for training descriptors to be matched with the Euclidean distance [45].

Siamese networks have also been successfully applied in remote-sensing image matching. Compared to natural imagery, the content of remote-sensing imagery is complex, and the uniqueness is low. In image matching tasks with small modal differences, scholars tend to use a Siamese network with shared parameters as the feature extractor or descriptor. The input of the network is two image patches centered on the key points, and the output of the network is the similarity of the two image patches [24], [26]. The two branches of the Siamese fully convolutional network (SFcNet) share the same infrastructure and the same weight [28]. Its two branches are connected by a convolution operation, and the strategy of maximizing the feature distance between positive and negative samples is adopted to output the similarity score between the two input image patches. This method has been tested on optical, near-infrared, thermal infrared, and synthetic aperture radar (SAR) images, but the matching accuracy for thermal infrared and visible images is not high.

For images with large modal differences, the strategy of fully sharing parameters in the Siamese network is inferior to that of the pseudo-Siamese network, which does not share parameters. The pseudo-Siamese network adopts different modal parameters for images with different modalities and can obtain a higher matching accuracy. This conclusion has been verified in many studies [29], [30]. In order to identify the corresponding patches in optical and SAR remote-sensing images, Hughes *et al.* [29] used a Siamese network with two independent weights, but the same convolution structure, to calculate the similarity between the SAR and optical image patches, to cope with the heterogeneity of the input images. Zhu *et al.* [30] adopted pseudo-Siamese network experimental multimodal image matching and trained the two-branch network independently, without sharing the network parameters. After the training, the two network branches are merged again as the input of the subsequent matching network. This kind of network design enables each branch of the network to retain its pure and important feature representation, which is conducive to improving the accuracy of the subsequent feature matching.

From the above analysis, we conclude that when the two images to be matched are of the same modality, the Siamese network with shared parameters can reduce the calculation cost because the images have similar and stable radiation information. When the two images to be matched are cross-modality heterogeneous data, a pseudo-Siamese network without shared parameters is required to learn modality-specific information. Our work is based on these findings.

III. Proposed CMM-Net Method

In this section, the cross-modality matching framework for thermal infrared and visible images is described in detail. In addition, a detection method for the key features is then

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introduced. Finally, the total loss function is proposed. Three kinds of loss function designs are illustrated, among which the MSD loss enlarges the interclass distance within the same modality, the CMC loss shortens the cross-modality intraclass distance, and the cross-modality triplet (CMT) loss is adopted to identify corresponding features.

A. Cross-Modality Image Matching Network

In this article, an end-to-end deep learning network is proposed for the cross-modality matching of thermal infrared and visible images, which we call CMM-Net. The framework makes use of both the unique information specific to the modality and the shared information independent of the modality, as shown in Fig. 1. In the shallow layers of CMM-Net, the modality-specific features in the thermal infrared domain and visible domain are individually extracted. In the deep layers of CMM-Net, the modality-shared layers are embedded for the extraction of modality-invariant features, which are beneficial to the cross-modality image matching.

1) Modality-Specific Network for Extracting Distinguishable Modality-Specific Features: Visible imagery describes the reflection of objects in the visible range, while thermal infrared imagery describes the thermal radiation capacity of the objects. Therefore, these image types are completely different in the expression of the visual features of the shallow layers. Due to the gap between features in the shallow layers between the different modalities, a modality-specific network is more advantageous in extracting distinguishable features. Therefore, a modality-specific network is adopted for both the thermal infrared imagery and visible imagery, and the parameters between them are unshared. Compared to the Siamese networks, CMM-Net pays more attention to extracting distinguishable modality-specific features in the shallow layers of the network, enabling the network to learn the modality-specific information between the thermal infrared and visible.

2) Modality-Shared Network for Embedding Modality-Invariant Features: Although the complex geometric and radiation variation between thermal infrared and visible images makes the appearance of the images show great differences, there is still some high-level semantic information shared between the different modalities, such as the structure and the spatial topological relations of objects. In the deep layers of the network, the features contain high-level semantic information. To learn and describe modality-shared features, the fully connected layers with shared parameters are embedded into the deep layers of CMM-Net. The embedded modality-shared layers contribute to capturing shared knowledge by creating links between the features of the thermal infrared and visible features. The modality-shared layers play an important role in the generation of modality-invariant features, which are beneficial to cross-modality thermal infrared and visible image matching.

B. Detection of Key Features

The feature map generated by the CNN is defined as F, $F \in \mathbb{R}^{h\times w\times c}$, where c represents the number of channels and $h \times w$ represents the size of the feature map in a channel. $F_{i,j,k}$ is the feature of position (i, j) on the kth channel of the 3-D tensor F, where $k = 1, \ldots, c$. For the purpose of back-propagation of the training process, a soft feature detection method is adopted for the detection of candidate key points, depending on the extremum points on the feature map in the spatial and channel dimensions.

First, in the spatial dimension, the soft local maximum $a_{i,j,k}$ is expressed as follows:

$$a_{i,j,k} = \frac{\exp\left(F_{i,j,k}\right)}{\sum_{(i',j')\in N(i,j)} \exp\left(F_{i',j',k}\right)} \tag{1}$$

where $N(i, j)$ represents the nine neighborhood pixels of pixel (i, j) in the image, including itself.

Second, in the channel dimension, a ratio-to-max of the channel-wise response is calculated by using the following:

$$b_{i,j,k} = \frac{F_{i,j,k}}{\max_{k'} F_{i,j,k'}}. \tag{2}$$

Next, to simultaneously consider the spatial and channel dimensions, the maximum product of the 2-D responses is calculated. Normalization is then conducted to obtain the soft detection score of the key features as follows:

$$d_{i,j} = \frac{\max_k \left(a_{i,j,k} \cdot b_{i,j,k}\right)}{\sum_{(i',j')} \max_k \left(a_{i,'j',k} \cdot b_{i',j',k}\right)}. \tag{3}$$

After the soft detection score is obtained, the feature vectors $F_{i,j}$ of all positions on the feature map are multiplied by their corresponding soft detection score $d_{i,j}$ to generate $d_{i,j} \cdot F_{i,j}$, which is used as the output of the final feature map.

C. Overall Loss Function for Cross-Modality Image Matching

By using shallow modality-specific layers and deep modality-shared layers in the network, we can achieve the goal of learning the cross-modality margin and extracting discriminative modality-invariant features. The addition of a Euclidean constraint to the modality-specific loss function can help align the cross-modality features of different domains. The overall loss function is expressed as follows:

$$L = L_{\text{CMT}} + \lambda_1 \left(L_{\text{MSD}}^{\text{TIR}} + L_{\text{MSD}}^{\text{VIS}}\right) + \lambda_2 L_{\text{CMC}} \tag{4}$$

where L_{CMT} is the CMT loss, $L_{\text{MSD}}^{\text{TIR}}$ is the MSD loss for the thermal infrared images, $L_{\text{MSD}}^{\text{VIS}}$ is the MSD loss for the visible images, and L_{CMC} is the CMC loss used to narrow the cross-modality gap. λ_1 and λ_2 are weighting coefficients of the loss functions. The influence of their values on the matching results is experimentally shown in the analysis of the weighting coefficients in Section V-B.

An intuitive schematic is shown in Fig. 2 to explain the role of the individual loss functions. $L_{\text{MSD}}^{\text{TIR}}$ and $L_{\text{MSD}}^{\text{VIS}}$ are used for expanding the interclass distance in the same modality, like pushing. L_{CMC} is adopted for reducing the intraclass distance in the different modalities, like pulling.

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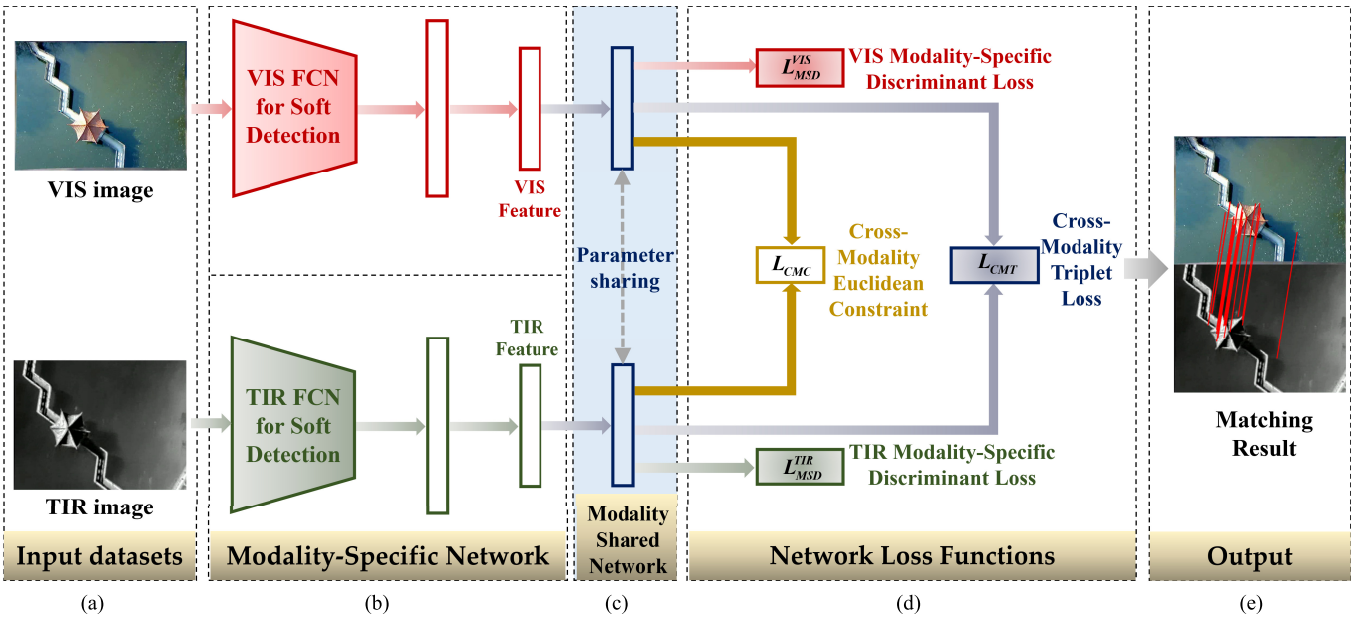


Fig. 1. Framework of CMM-Net. (a) Image pair input. (b) Modality-specific network for extracting distinguishable modality-specific features. (c) Modality-shared network for embedding modality-invariant features. (d) MSD loss is used for expanding the interclass distance in the same modality, the CMC loss is used for reducing the intraclass distance in different modalities, and the CMT loss is used for identifying corresponding features. (e) Image matching result output.

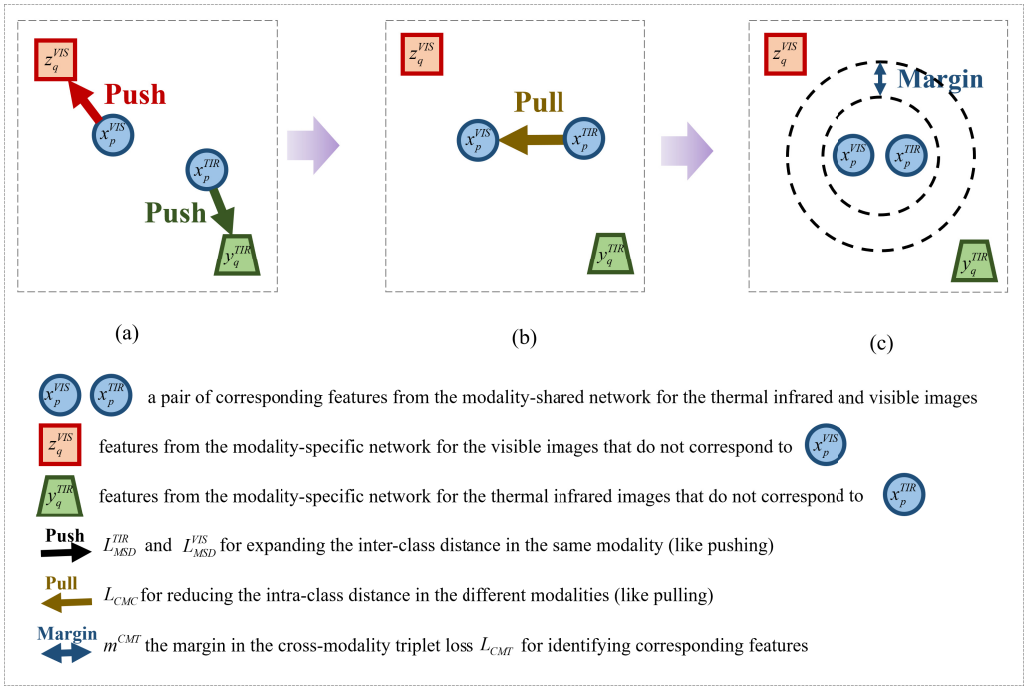


Fig. 2. Schematic of the loss functions. (a) MSD loss for expanding the interclass distance within the same modality. (b) CMC loss for reducing the intraclass distance between different modalities. (c) CMT loss for identifying corresponding features.

1) MSD Loss for Expanding the interClass Distance Within the Same Modality: In order to ensure that the network can be trained, positive samples and negative samples should be prepared. $\left(x_{\mathrm{p}}^{\mathrm{TIR}}, x_{\mathrm{p}}^{\mathrm{VIS}}\right)$, $p = 1, \ldots, N$, is defined as a pair of corresponding features output from the modality-shared layer, where $x_{\mathrm{p}}^{\mathrm{TIR}}$ belongs to the thermal infrared image, $x_{\mathrm{p}}^{\mathrm{VIS}}$ belongs to the visible image, and N is the number of corresponding features. Given conditions $q \neq p$ and $q = 1, \ldots, N$, $y_{\mathrm{q}}^{\mathrm{TIR}}$ represents the other key features output from the modality-specific network for the thermal infrared image that do not correspond to $x_{\mathrm{p}}^{\mathrm{VIS}}$, and $z_{\mathrm{q}}^{\mathrm{VIS}}$ represents the other key features output from the modality-specific network for the visible image that do not correspond to $x_{\mathrm{p}}^{\mathrm{TIR}}$. The corresponding features $x_{\mathrm{p}}^{\mathrm{TIR}}$ and $x_{\mathrm{p}}^{\mathrm{VIS}}$ are a pair of positive samples, while the non-corresponding

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features $\left(y_{\mathrm{q}}^{\mathrm{TIR}}\right.$ and $\left.x_{\mathrm{p}}^{\mathrm{TIR}}\right)$ and the non-corresponding features $\left(z_{\mathrm{q}}^{\mathrm{VIS}}\right.$ and $\left.x_{\mathrm{p}}^{\mathrm{VIS}}\right)$ are both negative samples.

To expand the interclass distance of the non-corresponding features in the same modality, the MSD loss function in the thermal infrared domain and visible domain is designed as follows:

$$L_{\mathrm{MSD}}^{\mathrm{TIR}} = \sum_{p=0}^{N} \left[m^{\mathrm{TIR}} - \left\| x_{\mathrm{p}}^{\mathrm{TIR}} - y_{\mathrm{q}}^{\mathrm{TIR}} \right\| \right]_{+} \tag{5}$$

$$L_{\mathrm{MSD}}^{\mathrm{VIS}} = \sum_{p=0}^{N} \left[m^{\mathrm{VIS}} - \left\| x_{\mathrm{p}}^{\mathrm{VIS}} - z_{\mathrm{q}}^{\mathrm{VIS}} \right\| \right]_{+} \tag{6}$$

where m^{TIR} and m^{VIS} are, respectively, the predefined margins in the thermal infrared modality and visible modality, which are used to increase the confidence of the corresponding feature matching. They are empirical values, so in this study, the optimal values were obtained through a large number of experiments. In addition, $[\cdot]_{+} = \max(\cdot, 0)$. The interclass distance of the non-corresponding features $\left(x_{\mathrm{p}}^{\mathrm{TIR}}, y_{\mathrm{q}}^{\mathrm{TIR}}\right)$ and $\left(x_{\mathrm{p}}^{\mathrm{VIS}}, z_{\mathrm{q}}^{\mathrm{VIS}}\right)$ in the different modalities needs to be larger than m^{TIR} and m^{VIS}. The key parameters m^{TIR} and m^{VIS} are further analyzed and discussed in the sensitivity analysis in Section V.

By means of maximizing the non-corresponding matching distance, the MSD loss functions $L_{\mathrm{MSD}}^{\mathrm{TIR}}$ and $L_{\mathrm{MSD}}^{\mathrm{VIS}}$ achieve the purpose of enlarging the interclass difference of the features.

2) CMC Loss for Reducing the intraClass Distance in the Different Modalities: The key to the image matching task is to find corresponding features, which requires high similarity of the corresponding features after the feature description. However, the visual representation of the different modality images for the same objects is likely to change, and thus corresponding features from different modalities may present completely different responses after learning from the network, which is harmful to cross-modality image matching. To narrow the cross-modality gap of the corresponding features, a cross-modality Euclidean distance constraint is adopted. The CMC loss L_{CMC} can be expressed as follows:

$$L_{\mathrm{CMC}} = \frac{1}{N} \sum_{p=1}^{N} \left\| x_{\mathrm{p}}^{\mathrm{TIR}} - x_{\mathrm{p}}^{\mathrm{VIS}} \right\|_2^2. \tag{7}$$

By minimizing the corresponding feature distance between the different modalities, the CMC loss makes the same features appear to have a more similar distribution after being extracted through the modality-shared layers.

When the gap between the different modalities is narrowed, the distribution of the visible features and thermal infrared features becomes less, resulting in smaller distances through the modality-shared layers in the different modalities.

3) CMT Loss for Identifying Corresponding Features: In order to identify the corresponding features of the cross-modality thermal infrared and visible images, a triplet loss function is used to minimize the corresponding feature distance while maximizing the distance of the non-corresponding features, where a predefined margin m^{CMT} is adopted to make the corresponding features more distinct from the non-corresponding features. The CMT loss L_{CMT} is expressed as follows:

$$L_{\mathrm{CMT}} = \sum_{p=0}^{N} \left[\left\| x_{\mathrm{p}}^{\mathrm{TIR}} - x_{\mathrm{p}}^{\mathrm{VIS}} \right\|_2^2 - \left\| x_{\mathrm{p}}^{\mathrm{TIR}} - x_{q}^{VIS} \right\|_2^2 + m^{\mathrm{CMT}} \right]_{+}. \tag{8}$$

The thermal infrared and visible features output from the modality-shared layers are controlled by the CMT loss, where the distance between corresponding features is less than that between non-corresponding features, so as to realize corresponding feature identification. m^{CMT} is also used to increase the confidence of the corresponding feature matching, so in this study, its optimal value was also obtained through a large number of experiments. The influence of the value of m^{CMT} on the matching results is analyzed in the parameter analysis in Section V-B.

IV. Experimental Results

In the comparison experiments with CMM-NET and the state-of-the-art image matching methods, thermal infrared and visible images from airborne and ground platforms were used as the experimental datasets to demonstrate the effectiveness of CMM-Net. In both qualitative and quantitative experiments, CMM-Net was compared with the handcrafted descriptors of SIFT [10] and position, scale, and orientation SIFT (PSO-SIFT) [46], as well as the two-branch network methods of SFcNet [28] and D2-Net [22], which are deep learning-based methods. In addition, this article explores how deep networks learn deep features, which have a certain receptive field, and are difficult to achieve at the subpixel level for the image matching task.

A. Experimental Settings

1) Implementation Details: At each branch of CMM-Net, ResNet [47] was used as the basic backbone of the network, but was truncated after the conv4_3 layer, which is used for key feature detection. The fully connected layers with unshared weights were then added to each network, outputting modality-specific features. Following the fully connected layers with unshared weights, the fully connected layers with shared weights were added to each branch to learn the modality-invariant features. All the experiments were performed on a computer with an NVIDIA Tesla P100 GPU accelerator and 16-GB GPU memory.

2) Training: For the UAV datasets, there are a total of 27 scenes, of which 23 were used for the training and four for the testing. For the computer vision center (CVC) datasets, a total of 140 scenes were used for the training and four scenes were used for the testing. To ensure that the test datasets were sufficient, rotation, inversion, cropping, and other operations were used to augment the data and then a training dataset of ten times the original image number was obtained. The ResNet network was pretrained on ImageNet [48]. Subsequently, the VIS_conv4_3 layer was used as the dense feature map of the visible image, and the TIR_conv4_3 layer was used as the dense feature map of the thermal infrared image. CMM-Net was then fine-tuned with the experimental dataset.

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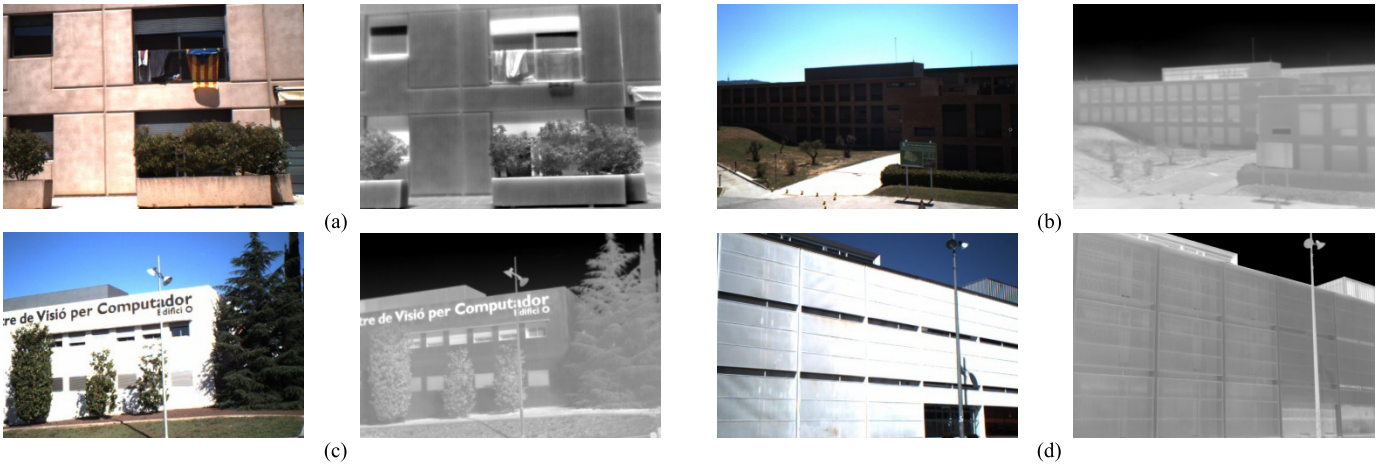


Fig. 3. Experimental test images from the CVC datasets. (a) CVC-1. (b) CVC-3. (c) CVC-2. (d) CVC-4.

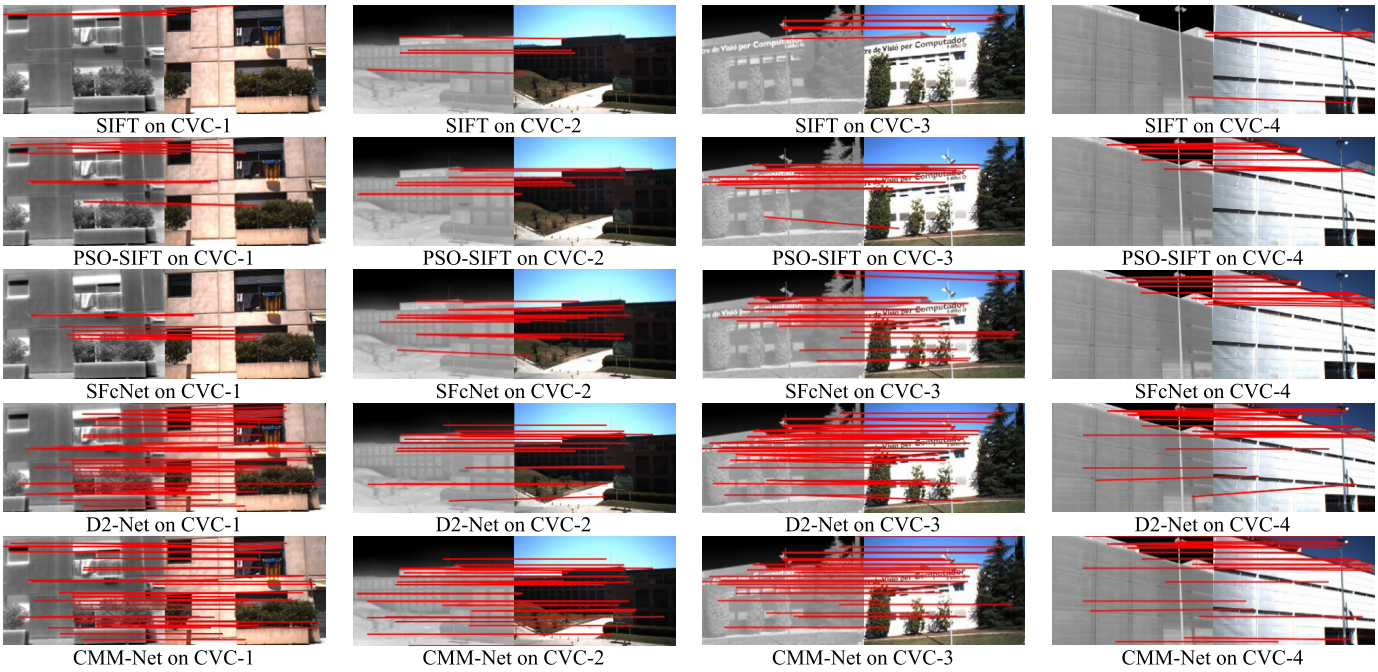


Fig. 4. Experimental results of CMM-Net and the state-of-the-art image matching methods for the CVC datasets.

For each corresponding point pair, we randomly selected a 256 × 256 image block centered on the corresponding point as the input. The parameters were optimized via random gradient descent, in which the momentum was 0.9, the weight attenuation was 10^{-4}, the initial learning rate was 10^{-3}, and the reduction was half per 1000 iterations. When the number of iterations reached 6000, the training was ended.

3) Testing: In order to obtain more initial key points and ensure a better positioning effect, the final pooling layer pool3 was changed from step size 2 to step size 1, and the convolution of conv4_1 to conv4_3 was changed to expansion convolution with the ratio of 2. Therefore, the acceptance field did not change.

4) Evaluation Metrics: To evaluate the different methods quantitatively, the number of points correctly matched (NCM), the ratio of correct points (TP), and the root-mean-square error (RMSE) are introduced as evaluation metrics. The matching point output of each algorithm was manually checked. When the error rate of a point was less than two pixels, it was considered to be a correct matching point. The ratio of correct points is the number of correct matches divided by the total number of matches (NM)

$$\mathrm{TP} = \frac{\mathrm{NCM}}{\mathrm{NM}} \tag{9}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{\mathrm{NCM}} \sum_{i=1}^{\mathrm{NCM}} \left[(x' - x)^2 + (y' - y)^2 \right]}. \tag{10}$$

To calculate the RMSE, 50 manual check points were selected for each set of images. The points matched by each algorithm were used to calculate the parameters of the image geometric affine transformation and then the RMSE was

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computed using the coordinates (x', y') calculated from the image transfer parameters and the coordinates (x, y) derived from the remote sensing image. The calculation equation for the RMSE is as follows:

B. Experiment 1: Ground Thermal Infrared and Visible Image Matching

1) Datasets: The experimental datasets for the ground thermal infrared and visible image matching were the CVC datasets [49], [50], which are publicly available. The CVC datasets consist of 144 pairs of thermal infrared and visible images of an urban outdoor scene, among which there is geometric distortion between the thermal infrared and visible images, as shown in Fig. 3. In the CVC datasets, the thermal infrared images were captured by a PathFindIR FLIR camera, with a resolution of 640 × 480. The visible images were taken by a Sony ICX084 camera, with a resolution of 534 × 426. The CVC datasets were obtained under different shooting times and imaging conditions, covering different scenes of urban areas. Therefore, the datasets are suitable for evaluating the robustness of the proposed CMM-Net method and the other image matching methods.

2) Qualitative Experiment: The CVC datasets were obtained by photographing objects in the urban scene with ground-mounted cameras. The photographed objects are relatively unique, with obvious contours and structures, and the imagery is less difficult to match than that of the UAV datasets. Fig. 4 shows the corresponding points matched by the proposed method and the state-of-the art methods.

As shown in Fig. 4, SIFT and PSO-SIFT, which are based on handcrafted descriptors, are poor at describing cross-modality heterogeneous data to mine higher levels of semantic information, so they fail completely on some thermal infrared and visible images. Similarly, SFcNet, which relies on handcrafted descriptors for the feature extraction, shows a poor image matching effect. CMM-Net focuses on enhancing the cross-modality image matching capabilities. CMM-Net obtains a large number of evenly distributed corresponding points. The proposed approach considers pushing negative samples into the same modality and adopts modality-specific measures for each modality to learn discriminative features. The matching results show that CMM-Net is suitable for the matching task of thermal infrared and visible images with large modal differences.

3) Quantitative Experiments: As defined before, TP, NM, and RMSE are adopted in the quantitative evaluation of the different methods for the ground CVC datasets. Tables I–III list the experimental results for the three metrics of the ground thermal infrared and visible image matching.

From Tables I–III, it can be seen that CMM-Net achieves the best precision and D2-Net ranks second. The performance of SFcNet lies in the middle level among the five compared methods. The SIFT and PSO-SIFT algorithms rely on significant texture features that differ greatly in the thermal infrared and visible images, so that they perform the worst of all. The nonlinear fitting ability of a deep CNN is helpful when identifying the cross-modality feature correspondence relationship, so that the deep learning-based methods are more conducive to the matching of thermal infrared and visible images. The receptive field generated by the CNN's multilayer convolution integrates spatial information to extract more reliable features and generate more distinguishable feature vectors. The modality-invariant feature representation has strong resistance to the nonlinear radiation distortion of images from different modalities. Similar results can be found for the methods that adopt cross-modality attitude quantities, such as SFcNet and D2-Net. D2-Net's intentionally delayed feature detection means that the detected features have higher-level semantic information. Applying the CMC loss can greatly reduce the corresponding feature distance between the thermal infrared domain and the visible domain, thereby improving the performance of the deep network and obtaining a higher precision for the matching between the thermal infrared and visible images. The analysis of the experimental results confirms the powerful cross-modality matching ability of CMM-Net.

C. Experiment 2: Airborne Thermal Infrared and Visible Image Matching

1) Datasets: The experimental datasets for the airborne thermal infrared and visible image matching were captured by UAVs. The UAV datasets [51], [52] consist of 27 pairs of thermal infrared and visible images taken by UAVs from an aerial view of urban areas, as shown in Fig. 5. For the thermal infrared images of the UAV datasets, the band range is 8–14 μm, and the size of the images is 640 × 480. For the visible images of the UAV datasets, they were photographed using a Dajiang (DJI) Matrice 100 platform, with a resolution of 5.0 μm × 5.2 μm, with the size of the images being 1280 × 720. The different imaging times and conditions resulted in complex geometric distortions between the thermal infrared images and the optical images, which were therefore suitable for testing the accuracy and effect of the proposed method.

2) Qualitative Experiment: The modal difference between the thermal infrared and visible images, as well as the abundant ground objects with different scales included in the images, made it very difficult to match the images of the UAV datasets. The qualitative experimental results of the proposed method and the other methods are shown in Fig. 6.

Due to the low flight altitude, small-scale coverage, and few unique features, the UAV-borne imagery has few distinguishable visual features. In view of the remote shooting of ground objects, the contours of the objects in the remote sensing imagery are very fuzzy, and the edges are difficult to accurately capture. The matching methods based on handcrafted descriptors (SIFT and PSO-SIFT) have difficulty in extracting significant texture structures, so that their matching effect is poor.

Although SFcNet adopts the structure of a Siamese network and aims to obtain more abstract high-level semantic information, the extraction of its initial key points relies on handcrafted descriptors, so that it is inadequate in the cross-modality matching task with complex heterogeneous data. D2-Net passes the extraction of key points to the deep network and applies the triplet marginal ranking loss

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TABLE I
NUMBER OF POINTS CORRECTLY MATCHED FOR THE CVC DATASETS

Dataset type	Dataset number	Method				
		SIFT [10]	PSO-SIFT [44]	SFcNet [26]	D2-Net [20]	CMM-Net
Ground CVC dataset	CVC-1	1	4	5	25	32
Ground CVC dataset	CVC-2	2	6	8	12	22
Ground CVC dataset	CVC-3	3	7	10	15	19
Ground CVC dataset	CVC-4	1	5	10	12	20

TABLE II
RATIO OF CORRECT POINTS FOR THE CVC DATASETS

Dataset type	Dataset number	Method				
		SIFT [10]	PSO-SIFT [44]	SFcNet [26]	D2-Net [20]	CMM-Net
Ground CVC dataset	CVC-1	25	33.33	78.65	90.42	92.62
Ground CVC dataset	CVC-2	50	50	86.34	91.43	94.36
Ground CVC dataset	CVC-3	50	66.67	88.46	92.88	90.31
Ground CVC dataset	CVC-4	50	75	89.22	90.67	93

TABLE III
ROOT-MEAN-SQUARE ERROR FOR THE CVC DATASETS

Dataset type	Dataset number	Method				
		SIFT [10]	PSO-SIFT [44]	SFcNet [26]	D2-Net [20]	CMM-Net
Ground CVC dataset	CVC-1	--	2.13	1.73	1.34	1.26
Ground CVC dataset	CVC-2	--	2.26	1.88	1.64	1.31
Ground CVC dataset	CVC-3	--	1.97	1.86	1.21	1.09
Ground CVC dataset	CVC-4	--	2.35	2.03	1.31	1.14



Fig. 5. Experimental test images from the UAV datasets. (a) UAV-1. (b) UAV-2. (c) UAV-3. (d) UAV-4.

under different modalities to match the corresponding features. However, the performance of D2-Net is worse than that of CMM-Net when the thermal infrared image is matched with the visible image since the cross-modality discriminative loss imposed by pushing negative samples enlarges the gap between the different modalities. Narrowing the gap is the key to generating modality-invariant features, which is of great assistance to the cross-modality image matching task.

3) Quantitative Experiments: For the quantitative experiments in airborne thermal infrared and visible image matching, the results for the three evaluation metrics are shown in Tables IV–VI.

According to Tables IV–VI, the proposed CMM-Net can obtain matching results on all the images of the UAV datasets with the highest accuracy. Furthermore, the UAV-borne images are more difficult to match due to the greater distance at which they are taken and the complexity of the objects

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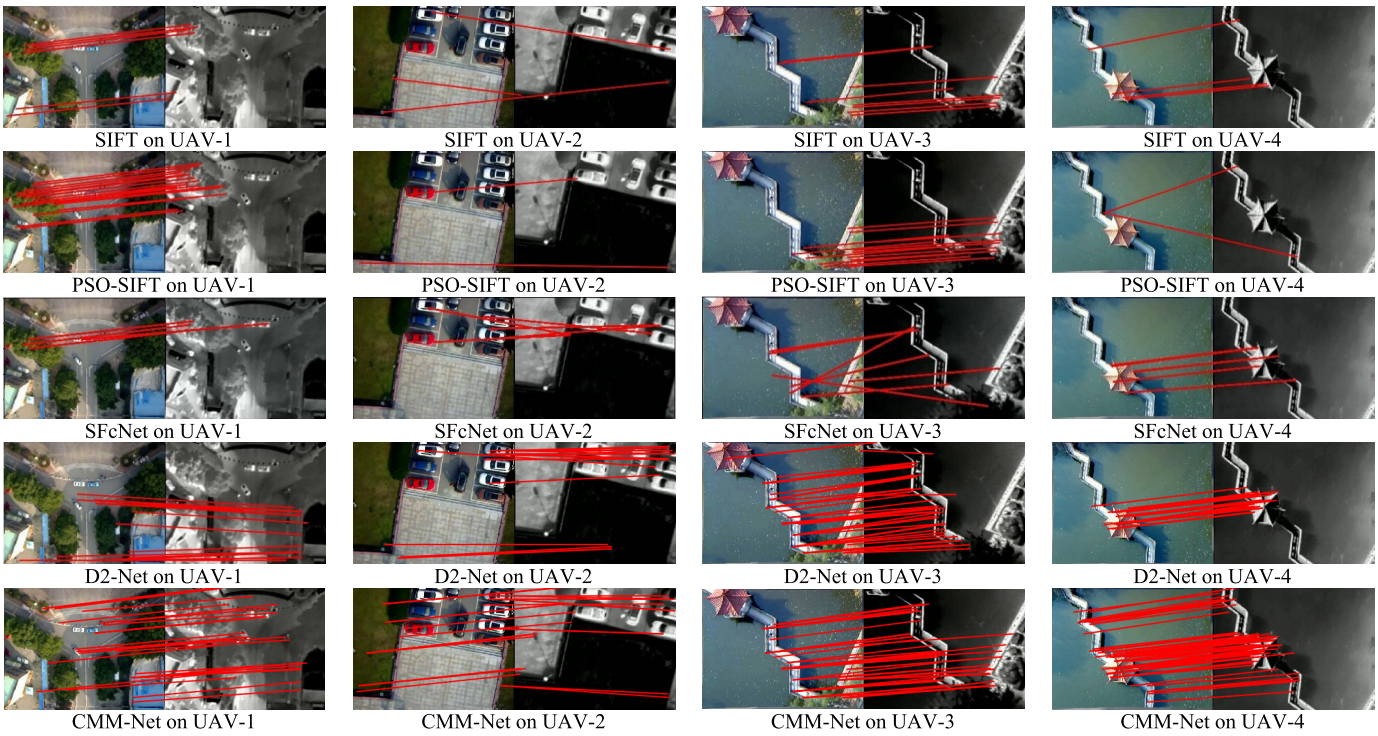


Fig. 6. Experimental results of CMM-Net and the state-of-the-art image matching methods for the UAV datasets.

TABLE IV
NUMBER OF POINTS CORRECTLY MATCHED FOR THE UAV DATASETS

Dataset type	Dataset number	Method				
		SIFT [10]	PSO-SIFT [44]	SFcNet [26]	D2-Net [20]	CMM-Net
Airborne UAV dataset	UAV-1	7	13	8	4	23
Airborne UAV dataset	UAV-2	--	--	--	6	15
Airborne UAV dataset	UAV-3	4	8	2	21	24
Airborne UAV dataset	UAV-4	1	1	4	11	28

TABLE V
RATIO OF CORRECT POINTS FOR THE UAV DATASETS

Dataset type	Dataset number	Method				
		SIFT [10]	PSO-SIFT [44]	SFcNet [26]	D2-Net [20]	CMM-Net
Airborne UAV dataset	UAV-1	66.67	76	79	85.42	91.5
Airborne UAV dataset	UAV-2	33.33	47.31	50	75	90.3
Airborne UAV dataset	UAV-3	71.43	73.21	78.76	84	93.29
Airborne UAV dataset	UAV-4	70.59	0	71.43	78.26	92.11

TABLE VI
ROOT-MEAN-SQUARE ERROR FOR THE UAV DATASETS

Dataset type	Dataset number	Method				
		SIFT [10]	PSO-SIFT [44]	SFcNet [26]	D2-Net [20]	CMM-Net
Airborne UAV dataset	UAV-1	2.33	1.87	1.69	--	0.97
Airborne UAV dataset	UAV-2	--	--	--	1.48	0.99
Airborne UAV dataset	UAV-3	--	1.98	--	1.41	0.96
Airborne UAV dataset	UAV-4	--	--	--	1.58	1.27

they contain, so that the difference in matching accuracy between SFcNet, D2-Net, and CMM-Net is more obvious than for the CVC datasets. Deep CNNs enable the extraction and description of key features to be embedded with rich spatial context information. The application of a non-linear activation function is conducive to the construction of cross-modal feature relations, so that the matching accuracy of D2-Net and CMM-Net is clearly better than that of SIFT and PSO-SIFT, which are based on handcrafted descriptors.

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TABLE VII
ABLATION STUDY WITH THE UAV DATASETS

Method	Coefficient	CMT	MSD	CMC	*TP*	*RMSE*
CMM-Net w/o MSD & CMC	$\lambda_1 = \lambda_2 = 0$	√			85.67	1.58
CMM-Net w/o CMC	$\lambda_2 = 0$	√	√		91.22	1.29
CMM-Net w/o MSD	$\lambda_1 = 0$	√		√	89.41	1.43
CMM-Net		√	√	√	93.77	1.15

From the quantitative results, it can be concluded that the unique network structure and well-designed loss function of CMM-Net successfully achieve cross-modality thermal infrared and visible image matching with a higher accuracy and stronger robustness than the state-of-the-art image matching methods.

V. FURTHER EVALUATION AND ANALYSIS

A. Ablation Study

Extensive ablation experiments were conducted to assess the effect of the important components and modules of the proposed network. Specifically, we evaluated the different components and configurations of CMM-Net by calculating the average of the TP and RMSE obtained from ten times experiments on the UAV datasets, as shown in Table VII.

In total, we considered four different network configuration schemes, each of which had different values of regularization coefficient relative to the overall loss function L.

1) Scheme 1 adopted a Siamese network with fully shared network parameters and structures as the two network structures matching the different modalities. The regularization coefficient of this scheme was $\lambda_1 = \lambda_2 = 0$.
2) Scheme 2 adopted a pseudo-Siamese network and MSD loss function to expand the interclass distance in the same modality, in which the pseudo-Siamese network was responsible for learning the network parameters specific to the modality. The regularization coefficient of this scheme was $\lambda_2 = 0$.
3) Scheme 3 adopted CMC loss on the basis of Scheme 2. Therefore, the regularization coefficient of the scheme was $\lambda_1 = 0$.
4) Scheme 4 was CMM-Net, with the regularization coefficients not set to zero. The experimental results of the four network configuration schemes are presented in Table VII.

From the experimental results, it can be found that the proposed network achieves remarkable results in the matching of thermal infrared and visible images. The performance of the network pretrained on ImageNet is better than learning from scratch. Both the modality-specific network structure and the embedding of the deep parameter-shared layers improve the accuracy. The shallow layers describe the modality-specific features, while the deep semantic layers capture the shared semantic information. Using the cross-modality Euclidean constraint significantly improves the performance of the deep features in the thermal infrared and visible image matching task. A detailed analysis follows.

1) Effectiveness of the MSD Loss: For modality-specific networks, the features of thermal infrared and visible images can be learned and extracted from the shallow network, the physical meaning behind the visual representation of the two types of images can be mined, and the distinguishable features can be generated in the thermal infrared domain and visible domain. We use the MSD loss function to learn the discriminative measure of the corresponding features in the thermal infrared and visible images, so as to improve the discriminative ability of the extracted features and ensure that the interclass variation is as much as possible. In the deep embedded modality-shared layers, the parameters are shared between the modalities, to mine shared knowledge and improve the modality invariance of the deep representation. Information from the modality-shared layers is propagated back to the modality-specific layers, which helps generate more differentiated corresponding features. From Table VII, with the MSD loss, it can be seen that the matching accuracy improves by 6.48%.

2) Effectiveness of the CMC Loss: The cross-modality Euclidean distance constraint reduces the large gap between the cross-modality thermal infrared and visible images by minimizing the intramodality distance. The CMC loss can effectively reduce the difference in visual representation between the two modalities and provides a basis for generating a modality-invariant representation, so as to improve the performance of the modality-specific framework. The experimental results listed in Table VII show that the CMC loss improves the accuracy by 4.37%.

B. Sensitivity Analysis

1) Weighting Coefficients: In this section, to analyze the key weighting coefficients in the overall loss function, we describe how we evaluated the weighting coefficients λ_1 and λ_2 of the modality-specific loss and the cross-modality Euclidean distance constraint. To change the value of the weighting coefficients is to magnify or weaken the role of each component in the network. Large values magnify the strength of the component, whereas small values reduce the strength of the component. The values with the highest accuracy obtained through many tests were set as the base values, that is, $\lambda_1 = 0.4$ and $\lambda_2 = 0.2$. Subsequently, four groups of experiments were carried out by changing the basic values of the weighting coefficients. See Fig. 7 for the detailed settings and results.

The weighting coefficients should be chosen reasonably to ensure that the corresponding features of the different modali-

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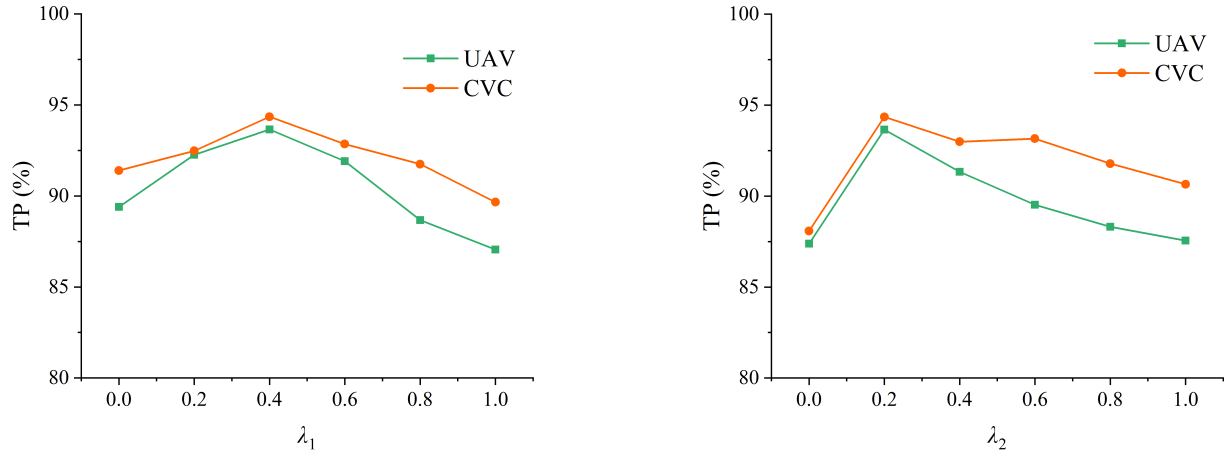


Fig. 7. Evaluation of the weighting coefficients λ_1 and λ_2 on the airborne UAV datasets and the ground CVC datasets (left: $\lambda_2 = 0.2$ and $\lambda_1 \in [0, 1]$; right: $\lambda_1 = 0.4$ and $\lambda_2 \in [0, 1]$), where the TP (%) is reported.

TABLE VIII
PARAMETER ANALYSIS ON THE UAV DATASETS

Margin parameter			Metric	
m^{TIR}	m^{VIS}	m^{CMT}	*TP*	*RMSE*
0.1	0.3	0.5	89.56	1.48
0.3	0.3	0.5	93.77	1.15
0.5	0.3	0.5	91.38	1.26
0.3	0.1	0.5	88.95	1.39
0.3	0.3	0.5	93.77	1.15
0.3	0.5	0.5	91.84	1.22
0.3	0.3	0.3	87.62	1.40
0.3	0.3	0.5	93.77	1.15
0.3	0.3	0.7	90.05	1.30

ties have high similarity and can be distinguished from features in the other modalities. When $\lambda_1 = 0.4$, the result has the highest precision, indicating that the effect of embedding the shared layer is similar to that of the convolutional layer specific to the modality. In contrast, when $\lambda_1 = 0.1$, the performance of the modality-specific network is suppressed. The network pays too much attention to minimizing the differences of the modalities, so that it cannot extract the thermal radiation information of the thermal infrared image and the texture information of the visible image, which results in a serious decline in the discriminative ability of the network. $\lambda_2 = 0.1$ indicates that the CMC loss is too small. At this point, the performance of a modality-specific network will be significantly reduced because the network cannot cross a large modality gap. After multiple comparison experiments, we suggest that the optimum value of the weighting coefficients is $\lambda_1 = 0.4$ and $\lambda_2 = 0.2$.

2) Parameter Analysis: Different margin parameters in the loss function result in different convergence. In this section, we describe the parameter analysis experiments undertaken to evaluate the margin parameters m^{TIR} and m^{VIS} in the MSD loss and the margin parameter m^{CMT} in the CMT loss. In particular, three sets of different parameters were evaluated $\big($0.1, 0.3, and 0.5 for m^{TIR}; 0.1, 0.3, and 0.5 for m^{VIS}; 0.3, 0.5, and 0.7 for $m^{\text{CMT}}\big)$. The parameter analysis results for the airborne UAV datasets are shown in Table VIII. From Table VIII, it can be seen that the value of the margin parameter has a great influence on the matching of thermal infrared and visible images. If the margin parameter is set too small, it is difficult to effectively distinguish the corresponding features from the non-corresponding features, thus bringing adverse effects to the matching results. In contrast, if the margin parameter is set too large, it makes it difficult to optimize the network. The recommended settings are $m^{\text{TIR}} = 0.3$, $m^{\text{VIS}} = 0.3$, and $m^{\text{CMT}} = 0.5$.

VI. CONCLUSION

Thermal infrared imagery can allow us to distinguish the targets from the background according to the thermal radiation difference and has the ability of being able to conduct imaging throughout the day and night, while visible imagery can provide high-resolution texture details in a manner consistent with the human visual system. After matching these two types of cross-modality images, their combined application and joint interpretation can make Earth observation methods more powerful. However, the image matching is challenging due to the heterogeneity of the data. In this article, after analyzing the problems inherent in the existing methods of cross-modality image matching, a network suitable for cross-modality image matching, called CMM-Net, has been proposed for learning a modality-invariant feature representation between thermal infrared and visible images. High-precision and robust matching results between the thermal infrared and visible images are realized by means of the modality-specific shallow layers and the modality-shared deep layers. Specifically, MSD loss is adopted to expand the interclass distance in the same modality, CMC loss is introduced to reduce the intraclass distance in the different modalities, and CMT loss is used to identify the corresponding features. In order to evaluate the accuracy and effectiveness of the proposed CMM-Net method compared to the other state-of-the-art methods in the matching of thermal infrared and visible images, two sets of data from airborne and

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ground platforms were adopted. From the quantitative results, it can be concluded that the unique network structure and well-designed loss function of CMM-Net can successfully achieve cross-modality thermal infrared and visible image matching with a higher accuracy and stronger robustness than the state-of-the-art image matching methods.

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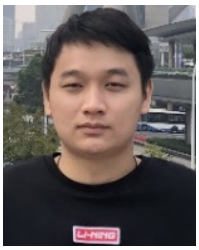
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